

CP476 Project Document

Course Registration Dashboard

Team Plan & User Stories

Team Plan

Roles

Permanent:

- **Scrum Master:** Nick (facilitates standups, unblocks team, owns sprint board hygiene)
- **Product Owner (rotating per sprint):** owns the milestone backlog, makes scope calls when the team is split

Functional (assigned for the term, can shift if workload demands):

- **Proposal and copy:** Manha, Isabel
- **Requirements and team coordination:** Nick
- **UX and wireframes:** Ayush, Alex, Naram
- **Data modeling and backend lead:** Matt
- **Repo, CI, GitHub Projects, branch protection:** Ahmed

Rotating (one person per sprint):

- **Demo lead:** prepares the milestone walkthrough
- **Scribe:** takes meeting minutes, posts to the wiki within 24 hours
- **QA lead:** verifies acceptance criteria on every story closed that sprint

Communication

- **Discord:** primary channel. Dedicated channels for #general, #standup, #frontend, #backend, #design, #blockers
- **GitHub Issues and Projects:** every unit of work is an issue with an assignee and a milestone
- **In-person:** working session on campus the weekend before each milestone deadline
- **Email:** only for prof and TA escalation

Meeting Cadence

- **Weekly sync:** Mondays, 60 minutes, in person. Agenda: review last week, plan this week, surface blockers
- **Async standup:** Discord #standup channel, post by 9pm Wednesday. Format: yesterday / today / blockers
- **Pre-milestone working session:** 48 to 72 hours before each deadline, minimum 2 hours, full team
- **Sprint retro:** 20 minutes at the end of each milestone, what worked, what did not, one change for next sprint

Sprint Structure

- Two-week sprints aligned to milestone deadlines
- Each sprint starts with planning (assign issues, set sprint goal) and ends with retro
- GitHub Projects Kanban with columns: Backlog, This Sprint, In Progress, In Review, Done
- Stories are tagged Easy, Medium, or Hard during planning to balance load across the team

Definition of Done

A task is Done only when all of the following are true:

- Acceptance criteria from the user story are met and verified by the QA lead
- Code is reviewed by at least one other team member via pull request
- PR is merged to main with no failing CI checks
- Tests are written for new logic (unit or integration, depending on layer)
- Documentation is updated where relevant (README, wiki, inline comments, or API notes)
- The work is demoed in the next standup or in the PR walkthrough
- The linked GitHub issue is closed and moved to the Done column
- For documentation deliverables (milestone PDFs, wiki entries): reviewed by two team members before submission

Frameworks & UI Tasks

- **Frameworks to be used:** React
- **UI TO DO (Alex):** Example course within course list table, Annotation for student view, Admin pages in Figma
- **UI TO DO (Ayush):** Login page admin feature, Annotations for admin page
- **UI TO DO (Naram):** Student view annotations, drop courses calendar, Design timetable, Add to cart

Requirements (User Stories)

US-01: Sign in to the dashboard

As a Laurier student

Requirement: sign in with university credentials

Goal: course plan and registration are tied to identity and protected

Acceptance Criteria:

- Sign-in screen accepts username and password
- Invalid credentials show an inline error and do not progress
- Successful sign-in lands on the dashboard home view
- Session persists across page reloads until explicit sign-out or 30 minutes of inactivity

US-02: Search the course catalog

As a student

Requirement: search courses by code, title, or keyword

Goal: candidates can be found without scrolling every department

Acceptance Criteria:

- Search input is visible on the catalog screen
- Results update within 1 second of typing (debounced)
- Partial matches in code (e.g. "CP3" returns CP311, CP312, CP317) and title (e.g. "algorithm") are returned
- Empty search shows the full catalog grouped by department
- No-results state displays a clear "no matches" message

US-03: Filter the course catalog

As a student

Requirement: filter courses by term, faculty, level, and time of day

Goal: options that fit the schedule and degree can be narrowed down

Acceptance Criteria:

- Filter panel exposes term, faculty, course level (100/200/300/400), and meeting time band (morning/afternoon/evening)
- Filters stack and combine
- Active filters appear as removable chips above the results
- A "clear all" action resets every filter
- Filter state survives a page reload

US-04: View course details

As a student

Requirement: open a course to see its description, prerequisites, sections, and meeting times

Goal: a decision can be made if it fits before committing

Acceptance Criteria:

- Clicking a course opens a detail view
- Detail view shows course code, title, description, credit weight, prerequisites, antirequisites, and all available sections (section number, instructor, days, time, room, seats remaining)
- Each section has an "add to plan" button
- A back action returns to the previous filtered or searched state without losing scroll position

US-05: Build a course plan

As a student

Requirement: add courses and specific sections to a plan before registering

Goal: experimentation is possible without committing

Acceptance Criteria:

- Adding a section moves it into the active plan
- Plan summary shows total credits, number of courses, and weekly hours
- The same course cannot be added twice (with a clear inline message)
- Items in the plan can be removed individually
- The plan persists across sessions for a signed-in user

US-06: See timetable conflicts visually

As a student

Requirement: a weekly grid view of the plan with conflicts highlighted

Goal: problems can be spotted immediately instead of guessing

Acceptance Criteria:

- Toggle between list view and weekly grid view of the plan
- Grid covers Monday through Friday across the time window the sections actually occupy
- Overlapping sections render with a red conflict outline and a tooltip naming the conflicting courses

- The plan summary shows a conflict count badge when count is greater than zero
- Resolving the conflict (drop or swap) clears the badge within 1 second

US-07: Submit a registration

As a student

Requirement: register for every section in the plan in one action

Goal: the workflow does not have to be repeated per course like the current system requires

Acceptance Criteria:

- The "register" action is enabled only when the plan has at least one section and no unresolved conflicts
- On submit, the system attempts to register each section in order
- Each section returns one of: registered, waitlisted, failed (with reason: no seats, prereq not met, time conflict, restriction)
- A confirmation screen lists per-section status
- Successfully registered courses move to the "current registration" view immediately

US-08: Drop or swap a registered course

As a student

Requirement: drop a registered course or swap it for a different section

Goal: the schedule can be adjusted without abandoning the rest of the registration

Acceptance Criteria:

- Each registered course exposes "drop" and "swap" actions
- Drop requires a confirmation modal before completing
- Swap opens a section picker filtered to the same course, with seats remaining shown per section
- After drop or swap, the current registration view reflects the change within 2 seconds
- An audit entry is written for each action and surfaced in the account history view

US-11: View remaining degree requirements

As a student

Requirement: see which degree requirements have been completed and which remain

Goal: courses are picked with a purpose, not at random

Acceptance Criteria:

- A "degree progress" view lists requirement categories (core, major, electives, breadth) with completion counts (e.g. 6/10 credits)
- Each category expands to show completed courses and a suggested list of courses that would satisfy remaining slots
- Courses in the current plan appear as "in progress" against the requirements they would satisfy
- The view updates after any registration change without a manual refresh

US-12: Use the dashboard on a phone

As a student

Requirement: the dashboard works on a phone

Goal: planning and registering on registration day is possible without needing a laptop

Acceptance Criteria:

- All screens render without horizontal scroll on a 375px wide viewport
- Touch targets are at least 44px tall
- The weekly grid view collapses into a daily list on viewports under 768px
- Critical actions (search, add to plan, register) remain reachable without nested menus

US-13: Add and manage courses (admin)

As an administrator

Requirement: add, edit, and remove courses in the catalog

Goal: the course offerings stay accurate each term

Acceptance Criteria:

- An admin-only management screen exposes create, edit, and delete actions for courses
- Creating a course requires code, title, description, credit weight, and at least one section
- Editing updates the catalog within 2 seconds and is reflected in student search results
- Deleting a course prompts a confirmation modal and warns if students have it in a plan or registration
- Non-admin users cannot access the management screen or its actions

US-14: Organize courses by term and semester

As an administrator

Requirement: assign courses to specific terms and semesters

Goal: students only see offerings relevant to the registration period

Acceptance Criteria:

- Each course/section can be tagged with a term (Fall/Winter/Spring) and academic year
- The catalog can be scoped to a single term, with a term selector at the top
- Courses without a term assignment are flagged in the admin view
- Changing a course's term moves it out of the prior term's catalog immediately

US-15: View a detailed course summary

As a student

Requirement: a consolidated summary view of a course's key information

Goal: it can be assessed at a glance without opening multiple sections

Acceptance Criteria:

- Summary surfaces code, title, credit weight, prerequisites, antirequisites, total sections, and seat availability across sections
- Summary is accessible from both the catalog list and the detail view
- Summary indicates whether prerequisites are satisfied
- Summary shows the earliest and latest meeting times across all sections

US-16: View registration in calendar format

As a student

Requirement: see registered courses laid out in a calendar

Goal: the week can be visualized as it will actually run

Acceptance Criteria:

- A calendar view renders current registrations across Monday through Friday
- Each entry shows course code, section, room, and time block
- The calendar reflects drops and swaps within 2 seconds
- The view distinguishes registered courses from plan-only (unregistered) courses

US-17: Filter courses by prerequisite satisfaction

As a student

Requirement: filter the catalog to courses whose prerequisites have been met

Goal: ineligible courses are avoided

Acceptance Criteria:

- A "prerequisites met" filter toggle is available in the filter panel
- When active, only eligible courses appear
- Courses filtered out by this toggle can be optionally shown with a "missing prereq" badge
- The filter combines with all other active filters

US-18: Filter courses by scheduling conflict

As a student

Requirement: filter out courses that conflict with the current plan

Goal: only sections that can actually be added are considered

Acceptance Criteria:

- A "no conflicts" filter toggle hides sections that overlap with anything in the active plan
- Conflict detection accounts for all sections currently in the plan
- The filter updates live as plan items are added or removed
- The filter combines with all other active filters

US-19: Filter courses by seat availability

As a student

Requirement: filter the catalog to only courses with open seats

Goal: time is not wasted on full sections

Acceptance Criteria:

- An "available seats only" filter toggle hides sections with zero seats remaining
- Seat counts refresh on filter application
- Sections that fill while displayed are visually flagged as full
- The filter combines with all other active filters

US-20: Filter courses by subject and course level

As a student

Requirement: filter courses by subject and level

Goal: focus can be kept on offerings relevant to the program and standing

Acceptance Criteria:

- The filter panel exposes a subject selector and a course level selector (100/200/300/400)
- Both filters support multiple selections
- Active subject and level filters appear as removable chips
- The filters combine and survive a page reload

US-21: Print the calendar of enrolled courses

As a student

Requirement: print or export the course calendar

Goal: an offline reference of the schedule is available

Acceptance Criteria:

- A "print" action generates a clean, paper-friendly layout of the calendar
- The printed output excludes navigation chrome and shows only the schedule grid
- Course code, section, time, and room appear in each block
- An export to PDF option is available alongside print

US-22: Auto-search for required co-requisites/labs on checkout error

As a student

Requirement: the system automatically finds required co-requisites or labs if registration fails for that reason

Goal: the error can be resolved without manual searching

Acceptance Criteria:

- When a registration fails due to a missing co-requisite or lab, the system identifies the required component
- A prompt offers to add the matching co-requisite/lab section to the plan
- If multiple eligible sections exist, a picker is shown
- Accepting the suggestion re-attempts the registration for both items

US-23: Auto-search Rate My Professor for a course's instructor

As a student

Requirement: see Rate My Professor information when viewing a course's instructor

Goal: instructor reputation can be factored into the choice

Acceptance Criteria:

- The course/section detail view shows a link or rating summary for the listed instructor
- The lookup is triggered automatically when the instructor is displayed
- If no match is found, the view shows a clear "no rating available" state
- The integration degrades gracefully and does not block the detail view from loading

US-24: Customize visible columns when registering

As a student

Requirement: set preferences for which columns are shown or hidden during registration

Goal: focus can be kept on the information that is cared about

Acceptance Criteria:

- A column-preferences control lets individual columns be toggled (e.g. instructor, room, seats, time)
- Preference changes apply immediately to the registration view
- Column preferences persist across sessions for a signed-in user
- A "reset to default" option restores the standard column set

US-25: Prompt login when unauthenticated

As an unauthenticated user

Requirement: be prompted to log in when attempting a protected action

Goal: guidance is provided to authenticate rather than hitting a dead end

Acceptance Criteria:

- Accessing a protected route or action while signed out triggers a login prompt
- After successful login, the original intended destination is returned to
- The prompt clearly states why login is required
- Public/read-only catalog browsing remains accessible without login (if in scope), otherwise the prompt appears on entry